

Asia Industrial Technology and Asia Emerging Robotics

Humanoid robots: The road toward attractive ROI in warehouses



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Investors have questioned the lack of practical applications for humanoid robots. As discussed in our previous report ([here](#)), we see killer applications emerge. In this note, we show that the payback period of humanoid robots for warehouse applications is around 3 years today and can be shortened to <1 year in both China and the western countries relatively soon with volume induced reduction of robot price. This will remove a critical hurdle for mass adoption.

Warehouse is a consensus field of humanoid early adoption. Even as humanoid technology remains imperfect and continues to improve, warehouse pilot projects have been actively underway (Exhibit 1). More than 20 humanoid players are active in this field (Exhibit 2), focusing on material handling and sorting, with partners, including Amazon, GXO, and SF Express, in e-commerce, logistics, and manufacturers' warehouses.

Why warehouse? We see three key reasons: 1) Humanoid robots cannot yet handle generalized tasks in open environments very well. Warehouse tasks (e.g. material handling, pick-and-place, sorting) are less complex than in manufacturing and households, and the structured environment makes a natural starting point (Exhibit 4 and Exhibit 5). 2) the automation demand of relatively standardized tasks is abundant in warehouses, making it easier for successful pilot projects to scale up; and 3) warehousing offers a clear pathway into broader logistics (Exhibit 3), especially **last-mile delivery**, which represents a highly attractive market.

Humanoid robots vs. existing warehouse automation. Warehouse automation has long been in demand (Exhibit 7 and Exhibit 8), leading to the deployment of various equipment (Exhibit 6) such as industrial robots, mobile robots (AGVs, AMRs), sorters, conveyors, AS/RS. However, adoption is heterogeneous, and many warehouses still operate at low automation levels and many workers still remain in even the most automated ones. Humanoid robots offer greater **flexibility** than conventional solutions to potentially fill in the automation gaps, especially in loading/unloading, sorting, and packaging.

ROI estimate. There are still technological and process hurdles for broader deployment, but cost is no longer a major concern. Given the different labor costs and robot prices (Exhibit 10 to Exhibit 14), we assess China and the U.S. separately (Exhibit 15 to Exhibit 20). Based on current industry progress (~75% of human worker efficiency + current robot prices), the payback period is about 3.3 years in China and 2.9 years in the U.S. If robotic efficiency reaches human parity, and unit prices fall by 50% to USD 25K in China and USD 75K in the U.S., the payback period will be reduced to 0.8 years in China and 0.7 years in the U.S., a very attractive ROI for large-scale adoption in warehouses.

DETAILS

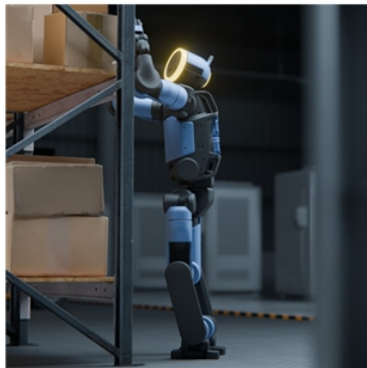
EXHIBIT 1: Humanoid robots work in warehouses: material handling and sorting



Agility Robotics



Appronik



Boston Dynamics



Figure



RobotEra



AgiBot

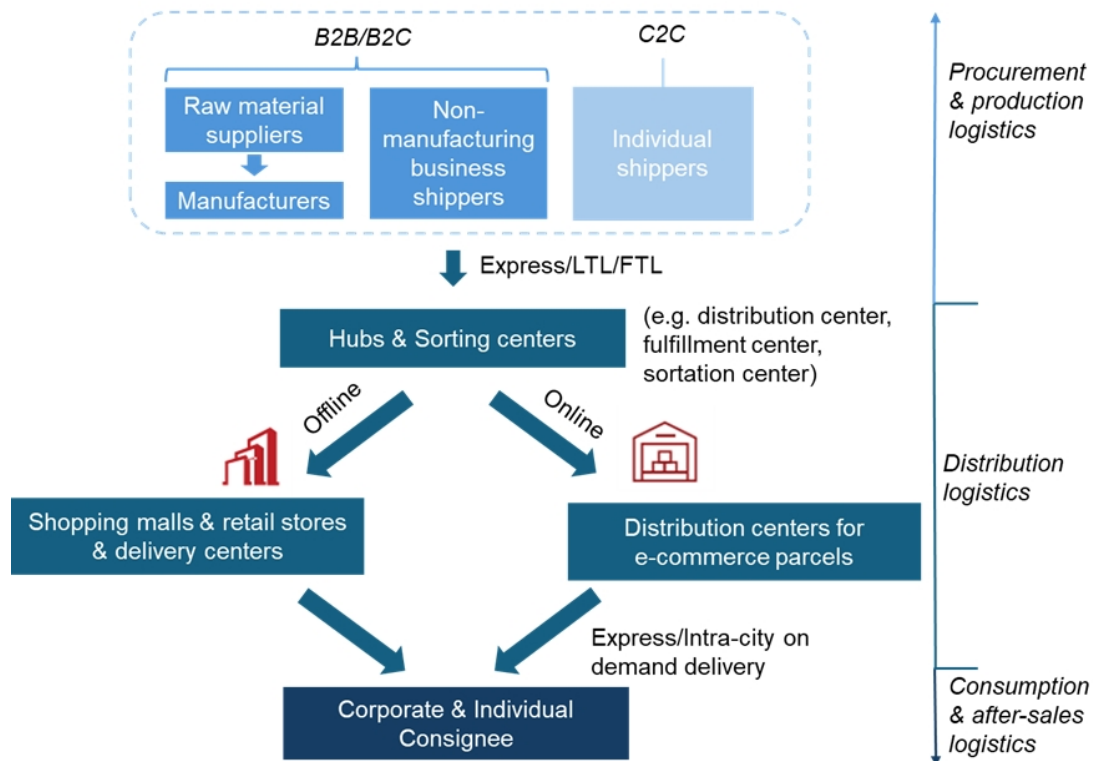
Source: Agility Robotics, Appronik, Boston Dynamics, Figure, RobotEra, AgiBot, Bernstein analysis

EXHIBIT 2: >20 humanoid players are exploring the potential of humanoid robots in warehouses

Company	Region	Ticker	Warehouse tasks		Customers for humanoid robots in warehouses
			Handling	Sorting	
Agility Robotics	USA	Private	√		GXO, Amazon, Toyota
Apptronik	USA	Private	√	√	Jabil
Boston Dynamics	USA	Subsidiary of Hyundai Motor Group	√		Amazon
Figure	USA	Private	√	√	
1X Technologies	USA	Private	√		
Reflex Robotics	USA	Private	√		GXO
Kinisi Robotics	USA	Private	√		
Humanoid	UK	Private	√		Siemens
Oversonic Robotics	Italy	Private	√		
Mentee Robotics	Israel	Private	√		
Rainbow Robotics	South Korea	277810.KS	√		CJ Logistics
VinMotion	Vietnam	Subsidiary of Vingroup	√		
AgiBot	China	Private	√	√	Damon
UBTech	China	9880.HK	√	√	BYD, Foxconn, Rossmann
RobotEra	China	Private	√	√	SF Express, China Post
Geek+	China	2590.HK	√		A global customer
Dobot	China	2432.HK	√		
Galbot	China	Private	√		
LimX Dynamics	China	Private	√		
PaXini	China	Private	√	√	
Kepler	China	Private	√		
Leju Robot	China	Private	√		
Standard Robots	China	Private	√		

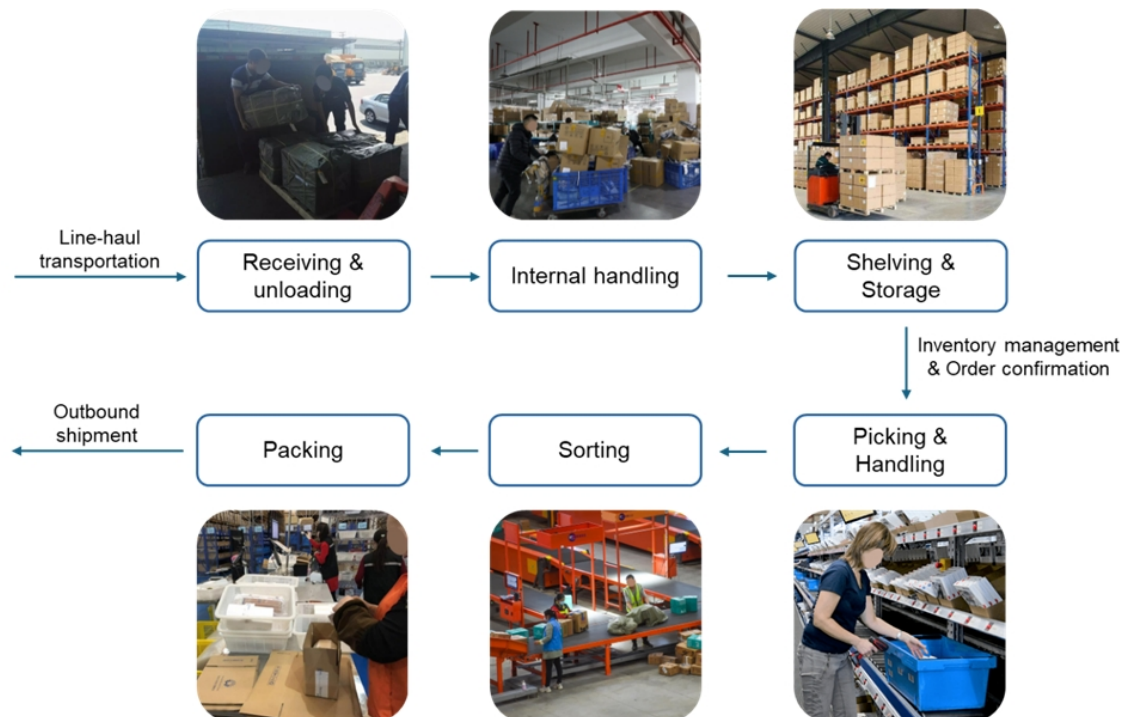
Source: Company websites, industry news, Bernstein analysis

EXHIBIT 3: Illustration of logistic process



Source: SF Holding Prospectus, Frost & Sullivan, Bernstein analysis

EXHIBIT 4: Workflow in warehouses



Source: Logclub, Logistics Business, Ntutong Logistics, Logistics salon, Changjiang Daily, Bernstein analysis

EXHIBIT 5: **Warehouses with different automation levels**

Low automation level



High automation level

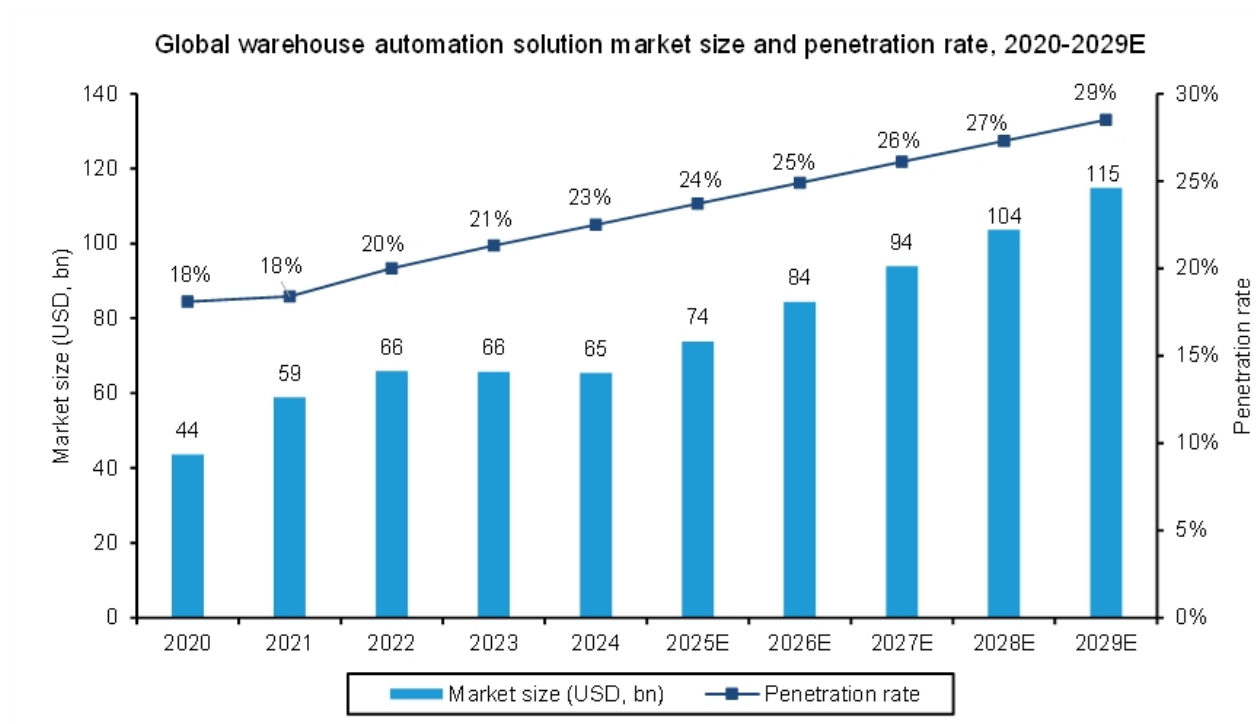
Source: The Break-down, Addverb, Bernstein analysis

EXHIBIT 6: **Typical solutions for warehouse automation**

	Solutions	Functions	Examples of typical equipment
Existing solutions	Mobile robots (incl. AMR, AGV)	Warehouse operations including goods or shelf handling , autonomous picking , and parcel sorting	 Underride  Bin-handling  Autonomous forklift  RollerTop
	Robotic arms	Repetitive tasks like loading/unloading , palletizing , and sorting	 Palletizing  Loading & unloading  Sorting
	Sorting systems	High-throughput and continuous sorting	 Cross-belt sorter  Linear sorter  Wheel sorter
	Automated storage & retrieval systems (AS/RS)	AS/RS, mainly for storage , are used for automatically storing and retrieving goods in a warehouse	 AS/RS  Stacker cranes  Shuttle robot
	Conveyor	Conveyors transports goods quickly and continuously within a warehouse	 Belt conveyor  Roller conveyor
Future solution	Humanoid robots	General-purpose warehouse worker	 Handling  Sorting  Packing

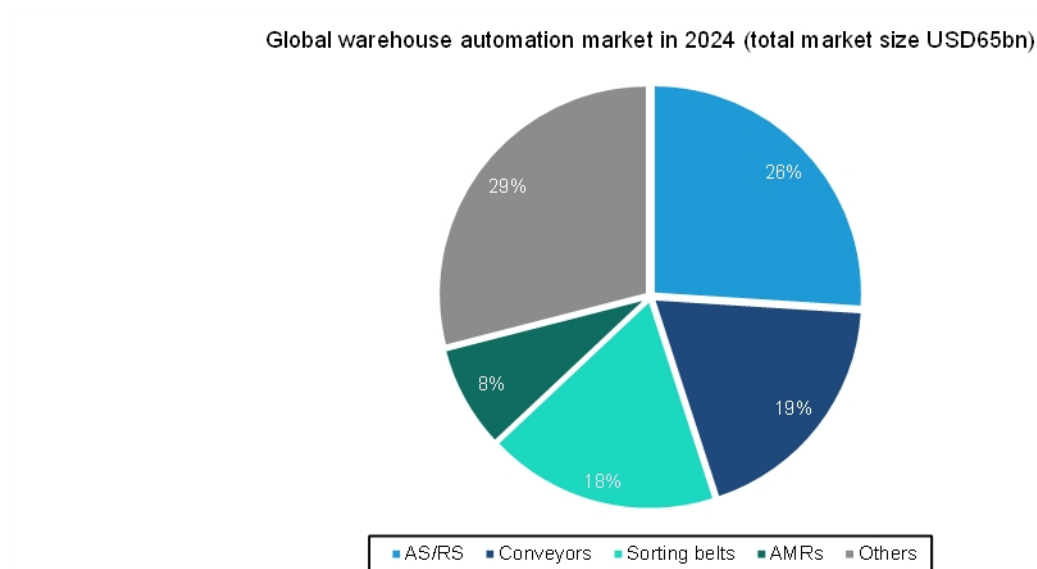
Note: AMR refers to autonomous mobile robot; AGV refers to automated guided vehicle; AS/RS refers to automated storage and retrieval system

Source: China Insights Consultancy, Suzhou Kuaijie, GoASRS, M Force Machinery, FANUC, Shenzhen Dongchang Automation Equipment, Nanjing Auto Electric, Bluesky Intelligence, Xiayun Machinery, Apptronik, Figure, AgiBot, Bernstein analysis

EXHIBIT 7: **Global warehouse automation solution market size and penetration rate**

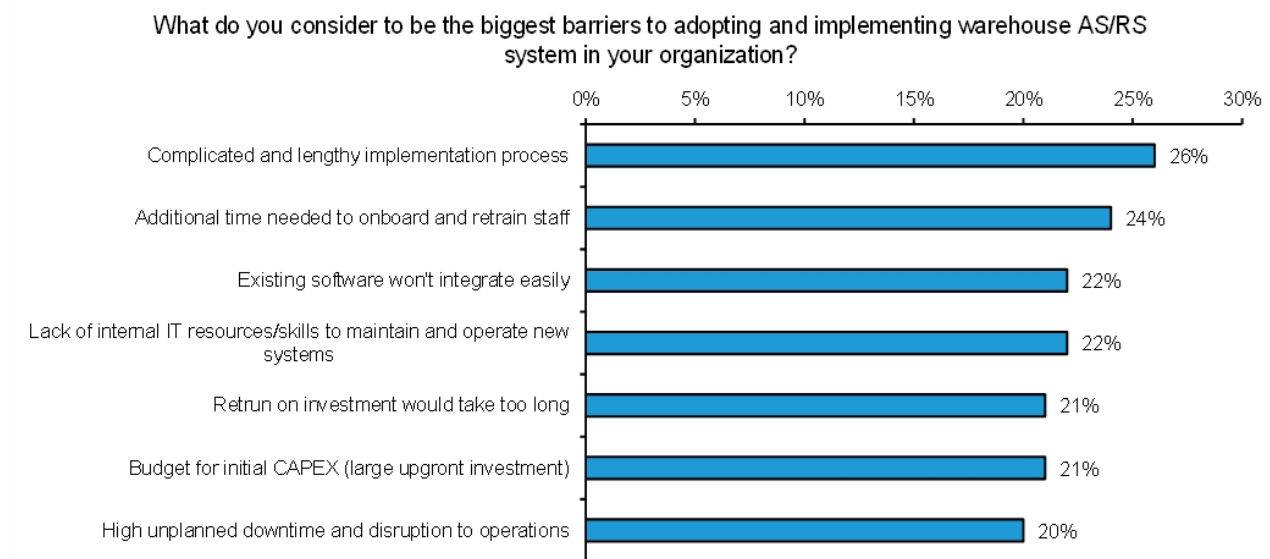
Note: The penetration rate of warehouse automation solutions is defined as the proportion of warehouses globally that have adopted any automation solution, relative to the total number of warehouses worldwide.

Source: Geek+ prospectus, Modern Materials Handling, CIC, Bernstein analysis and estimates

EXHIBIT 8: **Warehouse automation market breakdown by solutions in 2024**

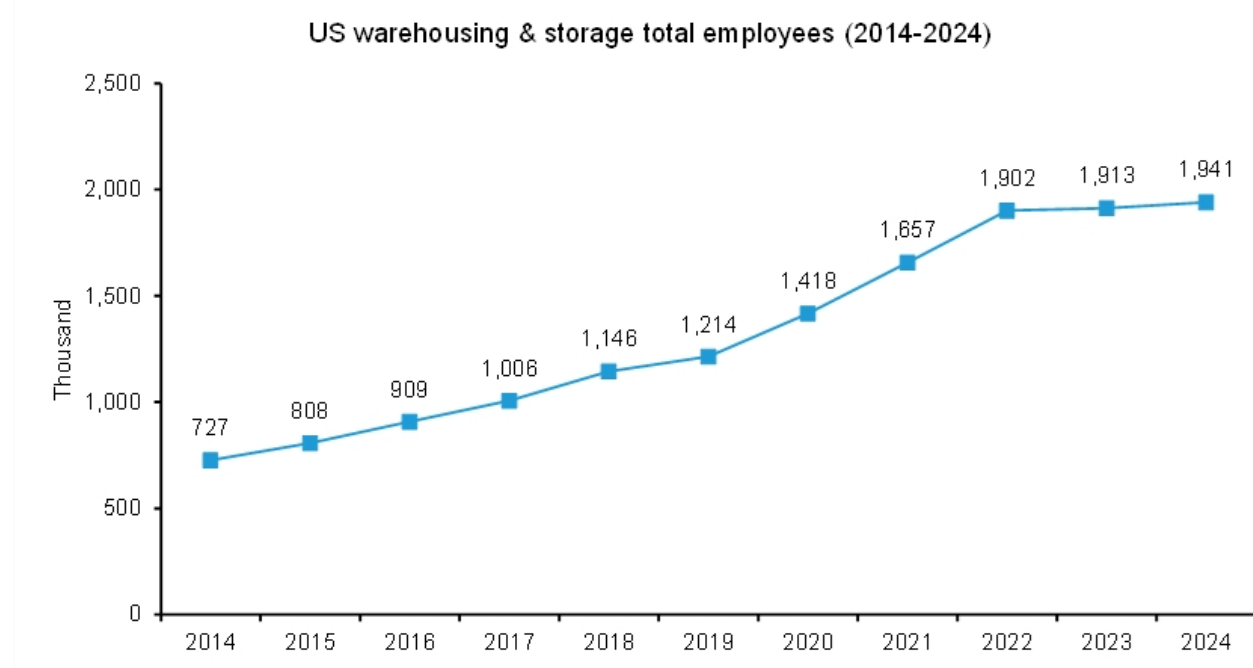
Note: AS/RS refers to automated storage and retrieval system; AMR refers to autonomous mobile robot.

Source: Geek+ prospectus, Bernstein analysis

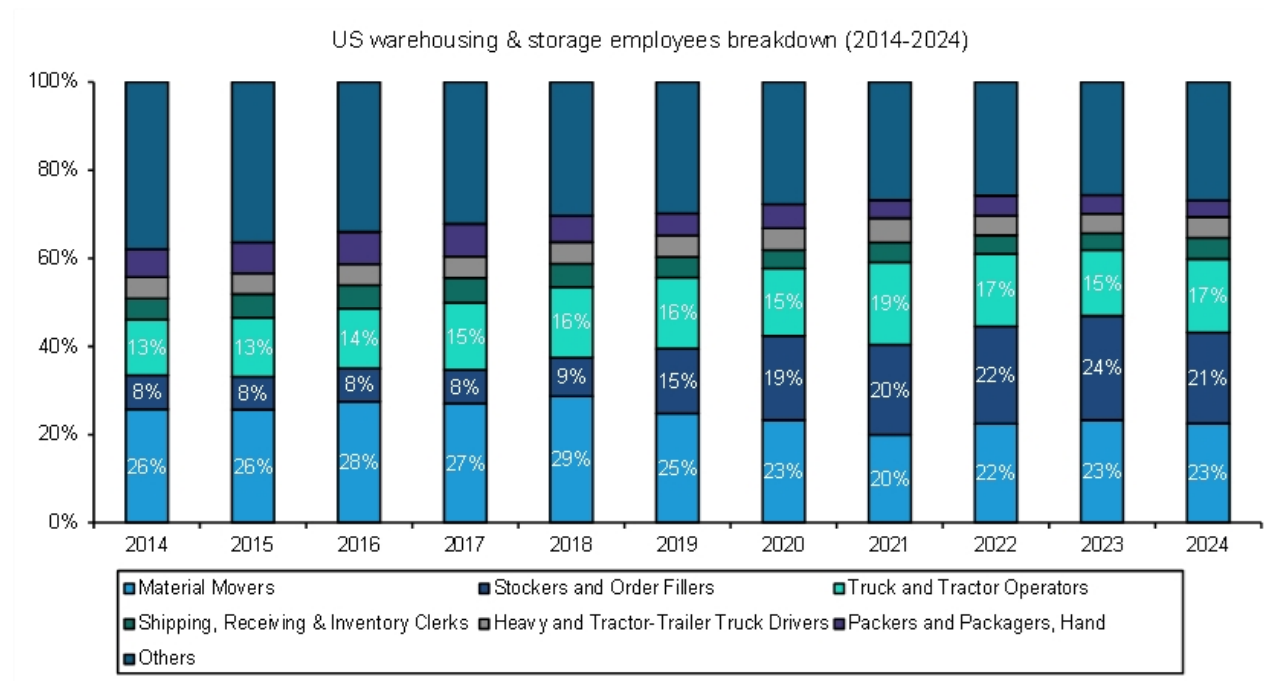
EXHIBIT 9: **Barriers to adopting and implementing warehouse automation**

Note: AS/RS refers to automated storage and retrieval system. This is a survey-based data, the sample size is 336. The survey interviewed a diverse group of decision-makers from across the global warehousing and fulfillment industry.

Source: AutoStore, Bernstein analysis

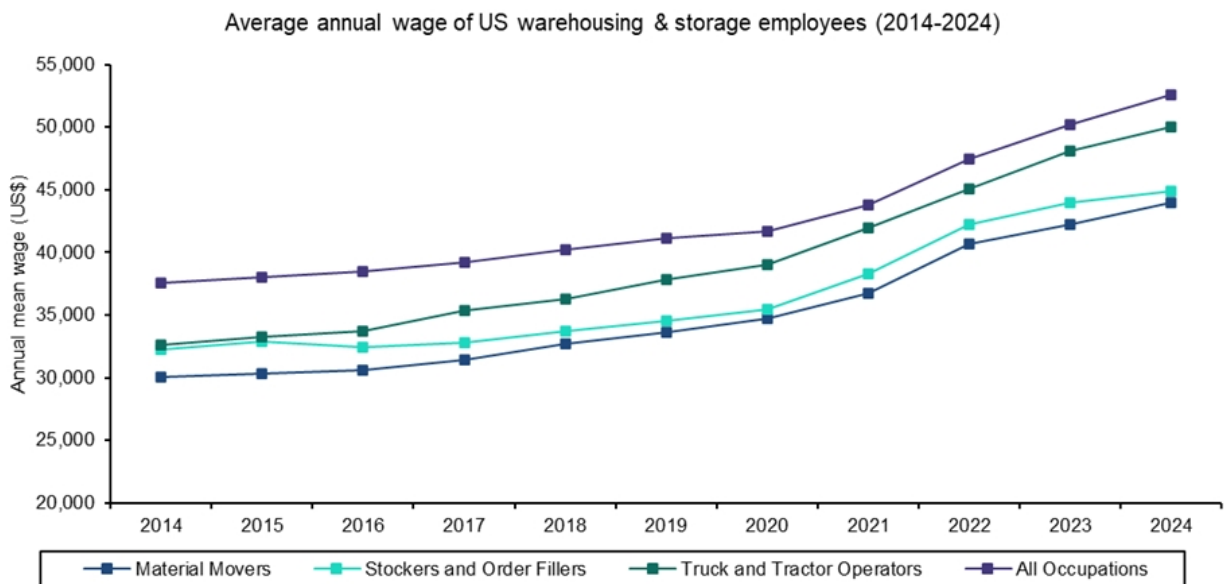
EXHIBIT 10: **US warehousing & storage labor demand has grown over the past decade**

Source: US Bureau of Labor Statistics (BLS), Bernstein analysis

EXHIBIT 11: The U.S. warehousing and storage workforce is shifting toward fulfillment-driven roles

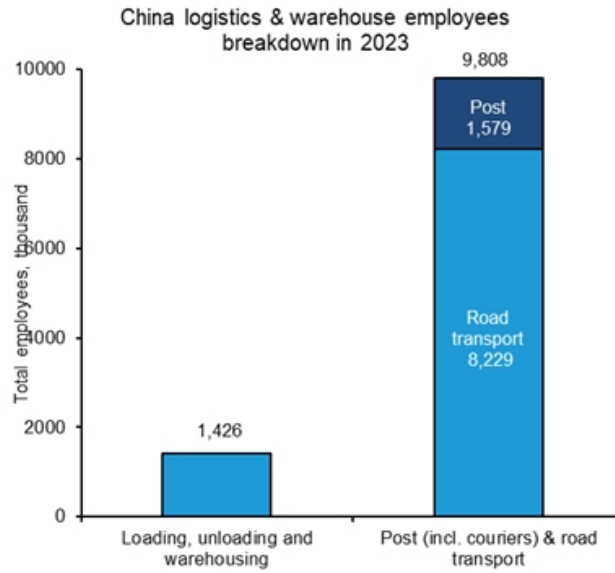
Note: Material Movers refer to "Laborers and Freight, Stock, and Material Movers, Hand", Stockers and Order Fillers refer to "Laborers receiving, storing, and issuing items from warehouse to fill shelves or customers' orders", Truck and Tractor Operators refer to "Industrial Truck and Tractor Operators"

Source: US Bureau of Labor Statistics (BLS), Bernstein analysis

EXHIBIT 12: The U.S.: Rising wages have increased labor costs in warehouses

Note: Material Movers refer to "Laborers and Freight, Stock, and Material Movers, Hand", Stockers and Order Fillers refer to "Laborers receiving, storing, and issuing items from warehouse to fill shelves or customers' orders", Truck and Tractor Operators refer to "Industrial Truck and Tractor Operators"

Source: US Bureau of Labor Statistics (BLS), Bernstein analysis

EXHIBIT 13: **China logistics and warehouse employees**

Note: Road transport includes urban public transport, highway passenger transport, road freight transport, and road transport support activities. Post (incl. couriers) includes postal services (basic services), express delivery services, and other delivery services.

Source: China National Bureau of Statistics, Bernstein analysis

EXHIBIT 14: **China logistics and warehouse employees' annual wages**

Note: Road transport includes urban public transport, highway passenger transport, road freight transport, and road transport support activities. Post (incl. couriers) includes postal services (basic services), express delivery services, and other delivery services.

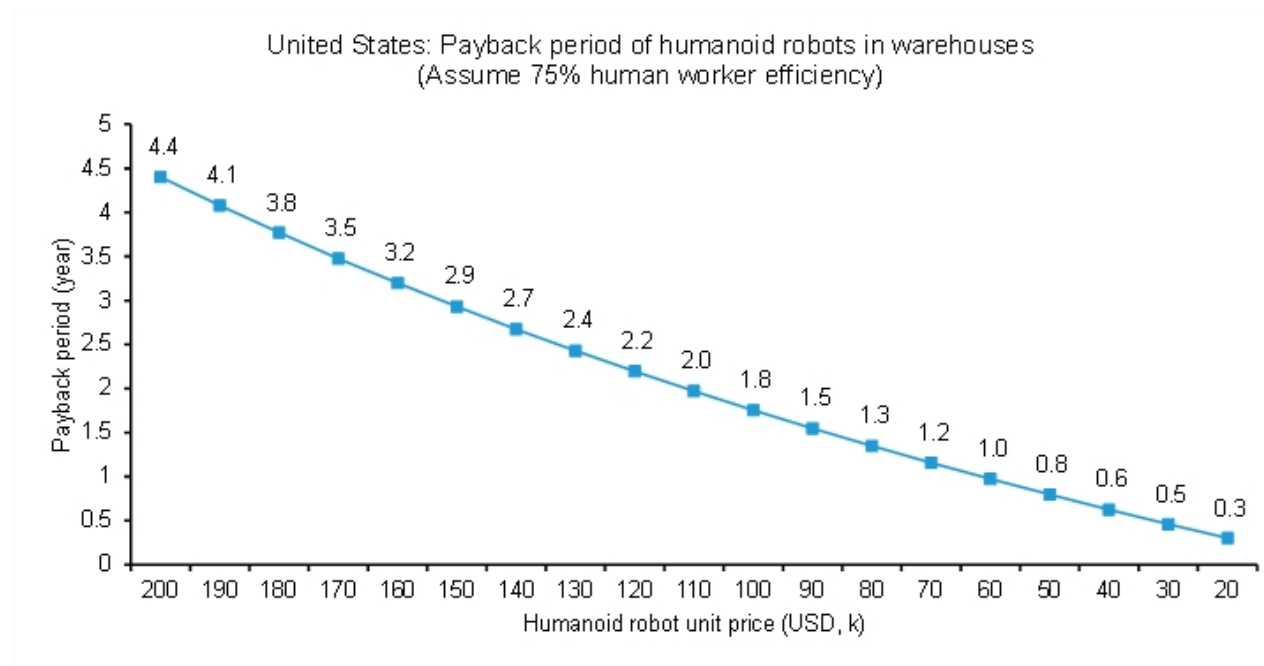
Source: China National Bureau of Statistics, Bernstein analysis

EXHIBIT 15: **Payback analysis of humanoid robots in the U.S. warehouses**

Estimated payback period of humanoid robots in the U.S. Warehouses				
Assume 1 humanoid robot to replace 0.75 warehouse worker (i.e. 75% human worker efficiency);				
Assume 1 human operator to look after 3 humanoid robots;				
Illustration of economics (every 2 humanoid robots in warehouse, all in USD)				
Upfront investment in humanoid robot solutions				
Upfront investment	# of humanoid robots	Robot unit price	Integration cost (% of robot price)	Sub-total
Procurement and system integration of humanoid robots	2	150,000	30%	390,000
Before replacement (only warehouse workers)				
Annual labor cost	# of workers	Unit labor cost	Implied shifts (relative to robot)	Sub-total
Warehouse workers	1.5	57,744	3.7	320,647
After replacement (humanoid robots + robot operators)				
Annual robot maintenance & electricity cost	# of humanoid robots	Annual robot cost (% of robot price)	Unit cost	Sub-total
Humanoid robots	2	15%	22,500	45,000
Annual labor cost	# of operators	Unit labor cost	Implied shifts (relative to robot)	Sub-total
Operators	0.7	57,744	3.7	142,510
Annual cost saved (excl. upfront humanoid robot investment)				133,137
Payback years				2.9

Assumptions	
Unit Labor cost	
Basic annual wage (USD, for both warehouse workers and robot operators)	44,419
Plus: welfare+miscellaneous (% of basic annual wage)	30%
Total annual wage (USD, for both warehouse workers and robot operators)	57,744
Operation mode & efficiency	
Humanoid robots' working efficiency relative to human workers	75%
# of humanoid robots looked after by 1 operator	3
Working time & shifts	
Working hours per day - human	8
Working hours per day - robot (with battery swap)	22
Implied shifts per day (human relative to robot)	2.8
Working days per year - human (i.e. 52 weeks * 5 working days per week)	260
Working days per year - robot (assume 15 days of maintenance per year)	350
Implied shifts per year (human relative to robot)	3.7

Source: US Bureau of Labor Statistics (BLS), Robozaps, Bernstein analysis and estimates

EXHIBIT 16: The U.S.: Price declines in humanoid robots will shorten the payback period

Source: US Bureau of Labor Statistics (BLS), Robozaps, Bernstein analysis and estimates

EXHIBIT 17: Sensitivity analysis of the payback period to humanoid robot prices and operational efficiency in the U.S.

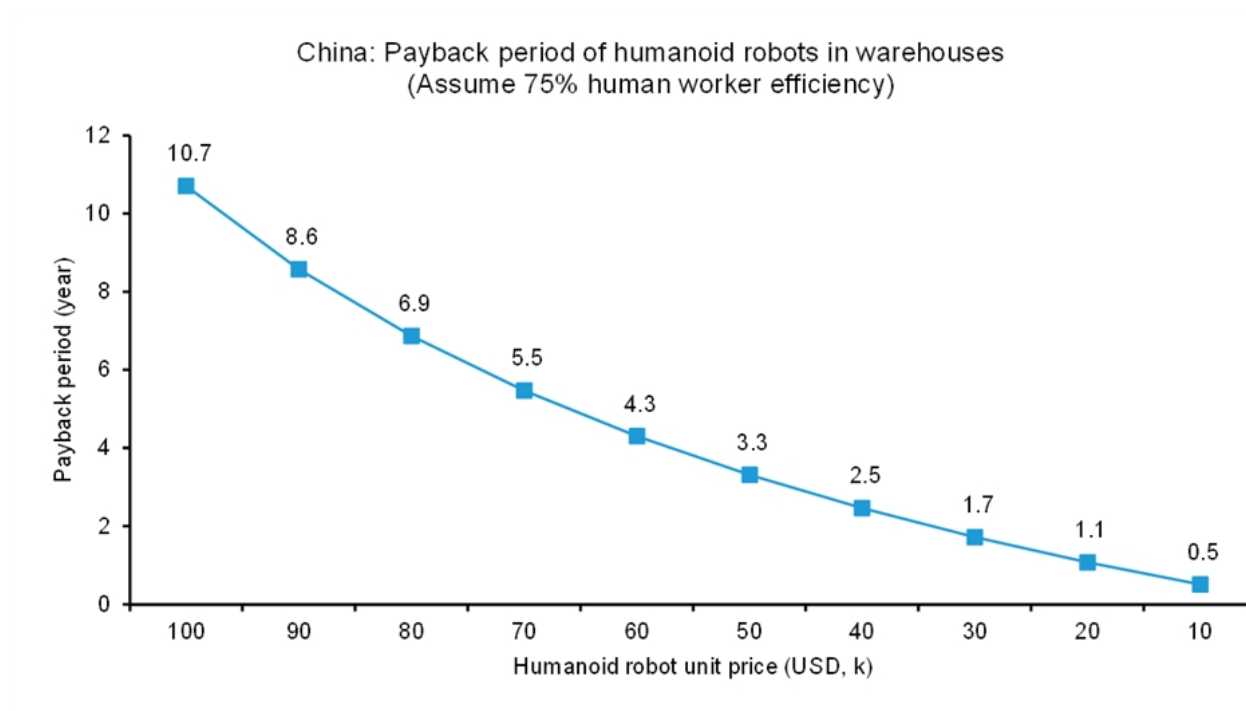
Payback period (year)		Humanoid robot unit price (USD, k)									
		200	180	160	140	120	100	80	60	40	20
Humanoid robots' working efficiency relative to human workers	65%	6.9	5.8	4.8	3.9	3.1	2.5	1.9	1.3	0.8	0.4
	70%	5.4	4.6	3.8	3.2	2.6	2.1	1.6	1.1	0.7	0.3
	75%	4.4	3.8	3.2	2.7	2.2	1.8	1.3	1.0	0.6	0.3
	80%	3.7	3.2	2.7	2.3	1.9	1.5	1.2	0.9	0.6	0.3
	85%	3.2	2.8	2.4	2.0	1.7	1.4	1.1	0.8	0.5	0.2
	90%	2.9	2.5	2.1	1.8	1.5	1.2	1.0	0.7	0.5	0.2
	95%	2.6	2.2	1.9	1.6	1.4	1.1	0.9	0.6	0.4	0.2
	100%	2.3	2.0	1.8	1.5	1.3	1.0	0.8	0.6	0.4	0.2
	105%	2.1	1.9	1.6	1.4	1.2	0.9	0.7	0.5	0.4	0.2
	110%	1.9	1.7	1.5	1.3	1.1	0.9	0.7	0.5	0.3	0.2
	115%	1.8	1.6	1.4	1.2	1.0	0.8	0.6	0.5	0.3	0.2
	120%	1.7	1.5	1.3	1.1	0.9	0.8	0.6	0.4	0.3	0.1

Source: US Bureau of Labor Statistics (BLS), Robozaps, Bernstein analysis and estimates

EXHIBIT 18: **Payback analysis of humanoid robots in China's warehouses**

Estimated payback period of humanoid robots in China's Warehouses				
Assume 1 humanoid robot to replace 0.75 warehouse worker (i.e. 75% human worker efficiency);				
Assume 1 human operator to look after 3 humanoid robots;				
Illustration of economics (every 2 humanoid robots in warehouse, all in USD)				
Upfront investment in humanoid robot solutions				
Upfront investment	# of humanoid robots	Robot unit price	Integration cost (% of robot price)	Sub-total
Procurement and system integration of humanoid robots	2	50,000	30%	130,000
Before replacement (only warehouse workers)				
Annual labor cost	# of workers	Unit labor cost	Implied shifts (relative to robot)	Sub-total
Warehouse workers	1.5	17,595	3.7	97,703
After replacement (humanoid robots + robot operators)				
Annual robot maintenance & electricity cost	# of humanoid robots	Annual robot cost (% of robot price)	Unit cost	Sub-total
Humanoid robots	2	15%	7,500	15,000
Annual labor cost	# of operators	Unit labor cost	Implied shifts (relative to robot)	Sub-total
Operators	0.7	17,595	3.7	43,424
Annual cost saved (excl. upfront humanoid robot investment)				39,280
Payback years				3.3
Assumptions				
Unit Labor cost				
Basic annual wage (USD, for both warehouse workers and robot operators)				14,663
Plus: welfare+miscellaneous (% of basic annual wage)				20%
Total annual wage (USD, for both warehouse workers and robot operators)				17,595
Operation mode & efficiency				
Humanoid robots' working efficiency relative to human workers				75%
# of humanoid robots looked after by 1 operator				3
Working time & shifts				
Working hours per day - human				8
Working hours per day - robot (with battery swap)				22
Implied shifts per day (human relative to robot)				2.8
Working days per year - human (i.e. 52 weeks*5 working days per week)				260
Working days per year - robot (assume 15 days of maintenance per year)				350
Implied shifts per year (human relative to robot)				3.7

Source: China National Bureau of Statistics, Robozaps, Bernstein analysis and estimates

EXHIBIT 19: China: Price declines in humanoid robots will shorten the payback period

Source: China National Bureau of Statistics, Robozaps, Bernstein analysis and estimates

EXHIBIT 20: Sensitivity analysis of the payback period to humanoid robot prices and operational efficiency in China

Payback period (year)		Humanoid robot unit price (USD, k)									
		100	90	80	70	60	50	40	30	20	10
Humanoid robots' working efficiency relative to human workers	65%	23.1	16.4	12.1	9.0	6.7	5.0	3.6	2.4	1.5	0.7
	70%	14.6	11.3	8.8	6.8	5.2	4.0	2.9	2.0	1.2	0.6
	75%	10.7	8.6	6.9	5.5	4.3	3.3	2.5	1.7	1.1	0.5
	80%	8.4	6.9	5.7	4.6	3.6	2.8	2.1	1.5	0.9	0.4
	85%	7.0	5.8	4.8	3.9	3.2	2.5	1.9	1.3	0.8	0.4
	90%	5.9	5.0	4.2	3.4	2.8	2.2	1.7	1.2	0.8	0.4
	95%	5.2	4.4	3.7	3.1	2.5	2.0	1.5	1.1	0.7	0.3
	100%	4.6	3.9	3.3	2.8	2.3	1.8	1.4	1.0	0.6	0.3
	105%	4.1	3.5	3.0	2.5	2.1	1.7	1.3	0.9	0.6	0.3
	110%	3.7	3.2	2.7	2.3	1.9	1.5	1.2	0.9	0.6	0.3
	115%	3.4	2.9	2.5	2.1	1.8	1.4	1.1	0.8	0.5	0.3
	120%	3.1	2.7	2.3	2.0	1.6	1.3	1.0	0.8	0.5	0.2

Source: China National Bureau of Statistics, Robozaps, Bernstein analysis and estimates

KOREA E-COMMERCE: COUPANG'S EARLY-INNINGS OF AUTOMATION

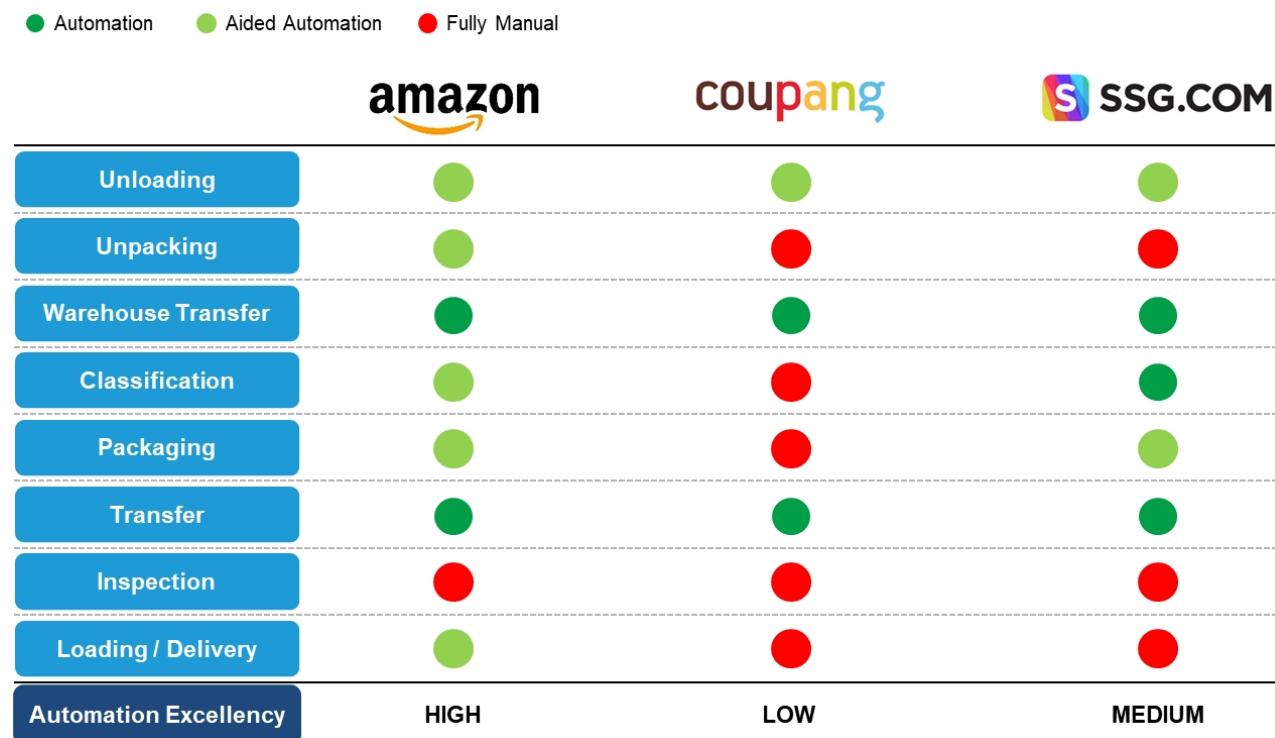
Automation spending reflects the front end of a multi-year build, with higher construction-in-progress, fixtures, and depreciation, but only modest productivity gains so far. Coupang's current initiatives focus on automating specific bottlenecks, with targeted investments such as Contoro Robotics fitting this pattern, rather than delivering fully robotic warehouses. Because warehouse automation rolls out in layers - from "AGVs (for goods transfer)" and "machine vision (for classification and sorting)" to "industrial robots (for palletizing and unloading)" and eventually "humanoid robots (in the 3 to 5 years, for material handling, loading/unloading, pick and place, delivery)" - efficiency gains are likely to come slowly and incrementally, with meaningful timing and impact extending well beyond the near term.

Initial design of Coupang's fulfillment centers are based on more prioritize flexible staging and manual handling over deep mechanization. The Seoul National University study indicates that Coupang's logistics centers look more like scaled, modernized versions of classic distribution warehouses than like fully reimagined, automation-centric facilities. The core spatial logic still follows conventional sequences of unloading, staging, movement, sorting, and loading, with automation added into segments

of the flow rather than defining the overall layout from first principles. This architectural evidence supports the view that Coupang's network, while sizable and capable, remains structurally closer to a traditional logistics system than to an Ocado-style or top-tier Amazon hub. The popular "robot factory" narrative therefore runs ahead of the physical reality on the ground.

Automation across Korea's e-commerce value chain still has a long runway. Last-mile delivery remains the biggest cost bucket, leaving clear margin upside if automation scales. But in Korea, execution risk is real - labor unions are likely to be the biggest constraint, and any attempt to automate last mile could quickly face pushback.

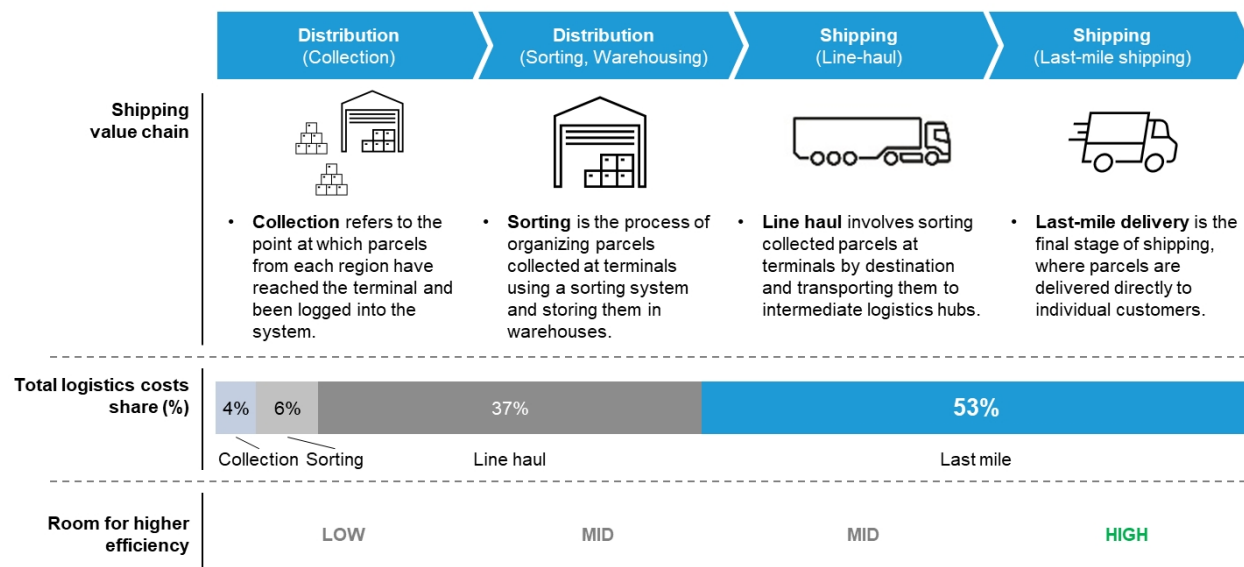
EXHIBIT 21: On-the-ground facility layouts tells Coupang's urban logistics sites favoring flexible staging and manual handling over heavy mechanization.



Source: Seoul National University, local press, Bernstein analysis.

EXHIBIT 22: As last-mile delivery represents the largest cost component in the value chain, we see potential margin upside for the broader Korean e-commerce industry as automation progresses, particularly if it extends into last-mile operations.

Korea E-Commerce: Margin expansion potential from last-mile delivery



Source: Business Insider Intelligence, KPMG, Bernstein estimates and analysis.

EXHIBIT 23: In Korea, though, labor unions will likely be the biggest roadblock - any push to automate last-mile delivery could quickly run into resistance from the workforce.

3 Different Types	Vehicle (Examples)	Contract Type	Vehicle Support	Car Insurance	Medical Insurance	Retirement pension	Sick Leaves	Labor Union Risks
Coupang Friend	 Coupang Owned Truck	Full-time • Contractor (< 2years) • Permanent (2+ years)	✓	✓	✓	✓	✓	✓
Coupang Flex	 Privately owned cars	• Self-employed • Part-timer (Time and parcel based)	✗	✗	✗	✗	✗	✗
Coupang Quick-Flex	 Commercial vehicles owned by individual drivers or 3PL company	• Self-employed • Part-timer (Time and parcel based)	✗	✗	✗	✗	✗	✓

Disclaimer: This report is based on publicly available information, third-party sources, and industry observations, and does not rely on direct input from Coupang's IR team. While efforts have been made to ensure accuracy, some information may be incomplete or subject to bias and should be interpreted accordingly.
Source: Expert interview, Truck driver community & forums, Truck driver interview, Bernstein analysis.

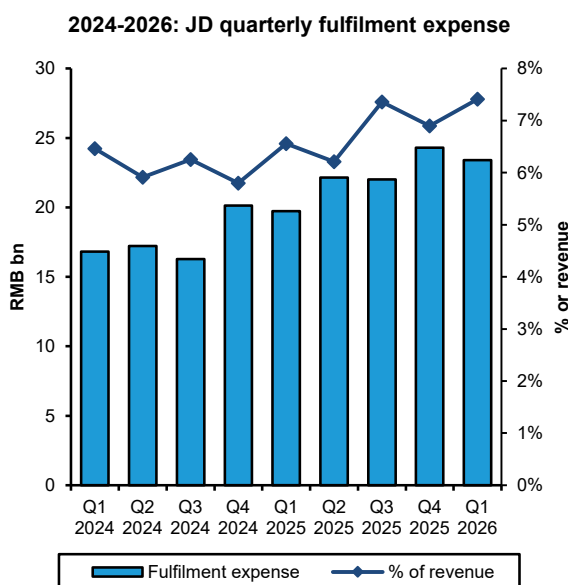
JD: WHAT CAN AUTOMATION DO FOR FULFILLMENT COSTS?

In China, JD represents one of the obvious users of automation. The company operates the largest first-party fulfillment network across our e-commerce coverage, and has spent the past year building out a network of front distribution warehouses to

support its expansion into quick commerce and same-day delivery. At the group level, JD's fulfillment costs represented 6.9% of revenue in the past four quarters ended March - we estimate this is around 4.1% of 1P GMV. Over the same period, fulfillment expenses represented 43% of JD's gross profit. As of the end of 2025, JD had a total of 777k employees, of which 643k were warehouse and delivery workers.

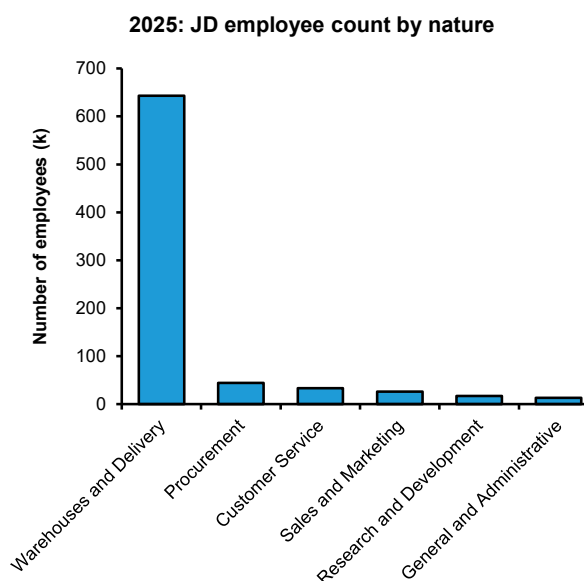
In our discussions, JD has spoken of continued investments in supply chain automation as one of the levers it has access to - in addition to SKU mix rationalisation and greater purchasing scale with key suppliers - to improve profitability over time. In an environment where AI investments (e.g. AI model development) have represented a notable driver of increased spend this year, JD stands out to us as being a potential beneficiary of physical AI growth - as robotics (humanoid or otherwise) technology make progress.

EXHIBIT 24: Fulfilment expenses have represented c. 7% of JD's revenue in recent years



Source: Corporate reports and Bernstein analysis.

EXHIBIT 25: 643k of JD's 777k employees at the end of 2025 were warehouse and delivery workers



Source: Corporate reports and Bernstein analysis.

INVESTMENT IMPLICATIONS

We rate Shuanghuan, Hesai, Tuopu, Inovance and Harmonic Drive Outperform; Sanhua Market-Perform; and Leader Drive Underperform.

We have **Outperform** rating on **JD**.

We have **Underperform** rating on **Coupang**.

BERNSTEIN TICKER TABLE

			21 May 2026	TTM		Reported EPS				Reported P/E (x)		
			Closing	Price	Rel.							
Ticker	Rating	Cur	Price	Target	Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
HSAL.US (Hesai)	O	USD	20.90	30.00	(44.0)%	CNY	2.92	3.31	5.27	48.7	42.9	26.9
2525.HK (Hesai)	O	HKD	166.90	238.00	NA	CNY	2.92	3.31	5.27	49.6	43.8	27.5
688017.CH (Leader Drive)	U	CNY	342.00	115.00	117.3%	CNY	0.69	0.83	0.96	493.9	410.0	355.7
601689.CH (Tuopu)	O	CNY	72.05	75.00	1.1%	CNY	1.61	1.81	2.36	44.8	39.8	30.6
2050.HK (Sanhua)	M	HKD	36.18	27.00	NA	CNY	1.03	0.97	1.11	30.5	32.3	28.4
002050.CH (Sanhua)	M	CNY	54.05	39.00	64.7%	CNY	1.03	0.97	1.11	52.5	55.6	48.8
002472.CH (Shuanghuan)	O	CNY	44.40	60.00	(8.2)%	CNY	1.50	1.70	2.00	29.6	26.1	22.2
9618.HK (JD)	O	HKD	121.70	155.00	(48.8)%	HKD	10.46	12.72	17.35	11.6	9.6	7.0
JD (JD)	O	USD	31.47	40.00	(34.5)%	CNY	18.15	22.08	30.12	11.8	9.7	7.1
CPNG (Coupang)	U	USD	15.73	15.00	(70.3)%	USD	0.10	(0.13)	0.30	158.7	(122.5)	52.6
300124.CH (Inovance)	O	CNY	78.99	82.00	(23.1)%	CNY	1.87	2.19	2.65	42.2	36.0	29.8
6324.JP (HDSI)	O	JPY	7,010.00	4,800.00	60.5%	JPY	36.57	15.51	57.16	191.7	451.9	122.6
ASIA			1,939.99									
SPX			7,473.47									
JPL			2,533.09									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

9618.HK, JD, CPNG estimate is Adjusted EPS; 9618.HK, JD, CPNG valuation is Adjusted P/E (x); 6324.JP base year is 2024;

Source: Bloomberg, Bernstein estimates and analysis.

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